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Fostering Consumer–Brand Relationships in Social Media Environments: The Role of Parasocial Interaction

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Abstract

As brands solidify their place in social media environments, consumers' expectations have amplified, thus spurring the development of technologies to assist with the engagement process. Understanding the ways in which brands can preserve the one-to-one characteristics and intimate relationship qualities offered by social media while still meeting consumer expectations amidst the escalating volume of interactions has become essential. Drawing on the communications literature, this research proposes that parasocial interaction (PSI) theory may be used as a theoretical lens for designing successful social media strategies. Three studies, using a multi-method approach, provide evidence of PSI's role in the development of positive relationship outcomes. Mediation analysis reveals that this sense of feeling connected with the brand goes beyond the interaction itself and drives increased feelings of loyalty intentions and willingness to provide information to the brand. Evidence from this research suggests that these effects may not hold when consumers are aware of the possibility that the brand's social media response may be automated. These findings offer marketers theoretical guidance for fostering relationships in social media environments.
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Keywords: Social media; Parasocial interaction; Online consumer behavior; Willingness to provide information; Brand loyalty; Interactivity; Openness in communication

> "We now ask the question, 'What will happen when a machine takes the part of A in this game?' Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, 'Can machines think?'"
>
> [– Alan M. Turing, Computing Machinery and Intelligence (1950)]

Introduction

Social media have altered the ways people communicate, collaborate, and connect with others and marketers have recognized its great capability for connecting with customers (Hennig-Thurau et al. 2010). Social media give marketers a means for direct interaction, which constitutes an ideal environment for creating brand communities (Scarpi 2010), establishing and reinforcing relationships, and for gaining a better understanding of consumers through netnographic research (Kozinets 2002). Yet, social media demand that marketers understand the environment if they are to avoid failures (Deighton and Kornfeld 2009; Hennig-Thurau et al. 2010; van Noort and Willemsen 2011), such as backlash which can reduce stock prices, damage reputations, create litigation costs, and even revenue loss (Butler 2011). Unlike static websites in the Web 1.0 era, the interactive nature of social media platforms developed in the Web 2.0 era has ultimately changed consumers' relationships with brands in these environments, even allowing them to become active players in the creation of brand stories (Gensler et al. 2013).

As social media usage increases, so do consumer expectations of brands, as evidenced by recent reports indicating that over one half of consumers now anticipate brand responses to consumer comments (Mickens 2012). As the number of consumers engaging with brands on these platforms steadily increases, firms are moving towards dedicated internal and external social media

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teams aided with software to assist automating the engagement process (Owyang 2012a, 2012b; Zebida 2012). While this hasn't quite reached the level of sophistication described in the opening quote, software that can be programmed to automatically and intelligently respond to consumer messages exists with the ability to integrate a number of custom variables to achieve a personalized interaction. As advances in machine learning are being applied to mine social media messages (Starbird, Muzny, and Palen 2012), it becomes realistic that it will become progressively difficult to distinguish machine from human response. As they decisively move towards automated engagement options (Owyang 2012a) it becomes essential to understand the ways in which marketers can preserve the intimate human relationship qualities offered by social media platforms while meeting consumer response expectations amidst the escalating volume of interactions.

In response, this paper pursues an empirical investigation of consumer–brand relationships on social media platforms by exploring the theoretical underpinnings that drive relationship development and the value they offer for companies. Drawing from the communications literature, parasocial interaction (PSI) theory is used to help explain a brand's success in developing strong ties with consumers through social media and provides insights on how to preserve intimate relationship feelings in light of the increasing movement towards response automation. Using this theoretical lens, two message components that transfer from traditional PSI environments to social media, perceived interactivity and openness in communication, are examined. Akin to a real-life relationship, this research proposes that PSI can result in positive relationship outcomes, specifically increased loyalty intentions and willingness to provide information.

To test these predictions, three studies are conducted using a multi-method approach. First, a survey with an online panel of adults explores consumers' active relationships with brands in social media environments. The results support the research premise that social media message cues (perceived interactivity and openness in communication) are two antecedents to the development of PSI. Moreover, feelings of PSI mediate the relationship between these message cues and the relationship outcome variables (loyalty and willingness to provide information). Ultimately, the sense of feeling connected to the brand through the interaction, not merely the interaction itself, drives these effects. Second, to further investigate this relationship and provide evidence of causality, the hypotheses are tested with an experimental design. The results from this experiment offer confirmation of the survey findings and provide causal evidence that message cues (perceived interactivity and openness in communication) can increase feelings of PSI, which in turn can increase loyalty and willingness to provide information. Lastly, Study 3 examines whether these effects might hold when participants become aware of the possibility that the brand's social media response may be automated. The results show that the effects observed in Studies 1 and 2 do not hold when the possibility for computer automation techniques have been made salient. Taken together, these findings offer marketers theoretical guidance for fostering relationships in social media environments.

Theoretical Background: Parasocial Interaction Theory

The concept of parasocial interaction emerged from the communications literature and offers an explanation of the development of consumer relationships with mass media, such as radio and television (Horton and Wohl 1956). PSI is described as an illusionary experience, such that consumers interact with personas (i.e., mediated representations of presenters, celebrities, or characters) as if they are present and engaged in a reciprocal relationship. In essence, people believe they are engaged in a direct two-way conversation, feeling as though a mediated other is talking directly to him or her (Houlberg 1984; Levy 1979; Rubin, Perse, and Powell 1985). PSI relationships can develop to the point where consumers begin to view mediated others as "real friends" (Stern, Russell, and Russell 2007). Feelings of PSI are nurtured through carefully constructed mechanisms, such as verbal and nonverbal interaction cues, and can carry over to subsequent encounters.

While some research presumes PSI is developed through multiple interactions, others provide evidence that the length of the relationship is not directly related to PSI (Perse and Rubin 1989) and that feelings of PSI can arise during initial exposures. While continued interactions should lead to enduring relationships and might strengthen these feelings, PSI can be created from signals in isolated interactions (Hartmann and Goldhoorn 2011). Additionally, while traditional PSI research focused on a viewer's relationship with a persona in broadcast media, recent research indicates that it may extend beyond these domains. For example, PSI might be cultivated through the design and presentation of information, such that it does not depend on the presence of a literal mediated personality such as a newscaster or actor (Hoerner 1999).

Extension of Parasocial Interaction Theory to Online Environments

In line with some recent applications of PSI to computer-mediated environments (Ballantine and Martin 2005; Hoerner 1999; Goldberg and Allen 2008), this research asserts that the development of PSI is not restricted to traditional mass media but can also be fostered through messages in an online environment that are designed to bring the viewer closer to a mediated persona, such as a brand or celebrity. While the Internet differs from traditional PSI environments (e.g., television and radio) in the fact that a direct two-way communication between an individual and the persona is technically possible, consumer-brand interactions on these sites oftentimes more closely mirror one-way conversations. For example, brand representatives typically base their responses in accordance with pre-approved scripts and response guidelines. Sometimes representatives are identifiable, but oftentimes this is not the case, leaving no clues to determine who is actually responding on behalf of the brand while supporting the perception that the message is coming directly from the brand (as opposed to employees on behalf of the brand). Moreover, in the case of multiple interactions with a brand, the brand responses are likely stemming from different employees yet appear to the

consumer as if they are stemming from a single respondent (i.e., the brand).

Moreover, the rise of social media has also brought the development of new technologies to help marketers manage these interactions without straining human resources. Automated software for social media now allows marketers to routinely post responses to consumer comments and can even scan message content to inform dynamically generated responses (Owyang 2012a, 2012b; Zebida 2012) or use geo-targeting to create seemingly personalized responses (Defren 2012). Regardless of whether or not the response originates from a live human being or a programmed script, this type of response can be more closely described as one-sided rather than two-sided communication, which is parallel to traditional PSI. Just as verbal and non-verbal message cues have been used to foster PSI in other mediums, it can be argued that message cues can be used to preserve the feelings of a two-way interaction between the persona and the brand, thus fostering PSI.

Accordingly, this research examines two message components that should transfer from traditional PSI settings to online environments. The first is the perceived interactivity of the persona with the viewer, which can be signaled through message cues that indicate responsiveness and listening. The second cue is openness in communication, which reflects the persona's self-disclosure, and can be signaled through the message content.

Fostering Parasocial Interaction

Interactivity

The marketing literature has established interactivity as an important feature of online environments (Song and Zinkhan 2008; Stewart and Pavlou 2002; Yadav and Varadarajan 2005) and has defined this term in a variety of ways. Despite being such an extensively researched construct, there is no general consensus of its definition (Johnson, Bruner, and Kumar 2006). Some researchers define interactivity according to the website's technical functionality (e.g., ability to navigate a site, provide feedback, and speed of the website); while others view it as perceptual variable (McMillan and Hwang 2002; Song and Zinkhan 2008).

In investigating the determinants of perceived interactivity, Song and Zinkhan (2008) found that both the speed of the response (speed) and the ability to communicate something that is related to a consumer's prior message (reaction) as being message features that can heighten perceived interactivity. In addition to identifying message cues that can increase perceived interactivity, they found evidence that perceived interactivity has a positive impact on perceptions of site effectiveness (i.e., satisfaction, loyalty, attitude toward the Web site, and site quality, repurchase behavior, and WOM).

The focus of this research follows Song and Zinkhan's (2008) conceptualization. Specifically, interactivity is defined as being dependent on the user's perception of taking part in a two-way communication with a mediated persona. In this context, the term interactivity does not focus on the technological features of the site, but on the content and cues within the message itself, which can be used to create an impression that the persona is listening to and interacting with the viewer in a timely fashion. This definition is aligned with the viewpoint that consumer perceptions of interactivity are more important than objectively defined features of a medium (Liu and Shrum 2002) and recognizes interactivity as a characteristic of the user (Steuer 1992).

In a traditional PSI context, such as television, various devices are used to support the development of perceived interactivity, including use of a subjective camera angle (i.e., the camera serves as the eyes of the audience), establishment of eye contact with viewers, and direct addresses of viewers (visually and verbally) (Auter 1992). Such efforts help the audience to feel as though it is being directly addressed, which intensifies feelings of PSI (Ballantine and Martin 2005).

As in typical social encounters, feelings of PSI should enhance a sense of mutual awareness and increase attention to the persona (Goffman 1983; Hartmann and Goldhoorn 2011). The viewer not only becomes aware of the persona but also develops a sense that the persona seems aware of the viewer. Perceived interactivity thus is contingent on creating an impression that the persona is listening and responding directly to the audience. This can be achieved through a message that contains elements of a direct two-way communication and through the timeliness of the reaction (McMillan and Hwang 2002; Song and Zinkhan 2008). Therefore,

H1a. PSI forms through message cues that increase perceived interactivity.

Openness

In communication, openness should increase feelings of PSI. Because PSI is described as akin to a friendship, the act of revealing information to a viewer should build intimacy and trust. Past PSI researchers describe this concept as "breaking the fourth wall," which means that the persona breaks away from his or her role to reveal information about him- or herself to the viewer (Auter 1992). This act of revealing gives the viewer the sense that he or she has gained inside information about aspects of the persona in this intimate setting (Meyrowitz 1986) and creates a feeling as if they know the persona on a much more personal level (Horton and Wohl 1956). In fact, viewers engaged in PSI relationships express desire for learning personal details about the persona, mimicking that of a real relationship (Stern, Russell, and Russell 2007). Furthermore, perceived self-disclosure fosters PSI through increasing feelings of intimacy and reducing feelings of uncertainty in the relationship (Perse and Rubin 1989), such that

H1b. PSI forms through message cues that signal openness in communication.

Outcomes of Parasocial Interaction

The two message components posited to foster PSI, interactivity and openness in communication, are also noted as important components in the relationship marketing literature for building trust and the development of interpersonal relationships. The relationship marketing literature supports the notion that

timely and relevant communication is a major precursor for the development of perceptions of trust and encourages the expansion of committed relationships, which increases both loyalty intentions and cooperation (Morgan and Hunt 1994). In addition to the importance of two-way communication, the perception of openness in communication is critical for establishing strong relationships (Anderson and Weitz 1992). Moreover, past research has established the use of openness and receptivity in communication as important trust building mechanisms for cultivating online relationships between buyers and sellers (Saini and Johnson 2005). PSI experiences are described as resembling interpersonal relationships, so much so that individuals "feel that they know and understand the persona in the same intimate way they know and understand flesh-and-blood friends" (Perse and Rubin 1989, p 60) and that the voluntary nature and ability to provide companionship can cultivate these strong bonds (Perse and Rubin 1989).

Therefore, outcomes of PSI experiences should be similar to those of "real" interpersonal relationships. Studies examining the effects of PSI support this notion. For example, research on PSI and soap opera television viewing reveals that viewers share some of the felt connections with the personas as they do with their real-life friends (Stern, Russell, and Russell 2007). PSI can increase engagement (Grant, Guthrie, and Ball-Rokeach 1991; Rubin, Perse, and Powell 1985) and those engaged in PSI strive to affirm their relationship with the mediated persona (Grant, Guthrie, and Ball-Rokeach 1991; Horton and Wohl 1956) through behaviors such as increased viewing and purchasing from the programs to which they are attached (Hofstetter and Gianos 1997; Levy 1979; Park and Lennon 2004; Rubin and Step 2000; Skumanich and Kintsfather 1998).

Researchers have also found that PSI is a better predictor of television viewership than many other behavioral measures, indicating that PSI may be a more important viewing motivation than that of the program content itself (Conway and Rubin 1991). Feelings of PSI also relate to satisfaction with television shopping experiences (Lim and Kim 2011) and increase enjoyment and commitment to social norms (Hartmann and Goldhoorn 2011).

Furthermore, messages in mediated interactions that resemble interpersonal communications can increase message credibility and persuasiveness (Beniger 1987). By increasing perceptions of credibility, PSI can alter attitudes and behaviors (Rubin and Step 2000), which likely stems from the active and involving character of high PSI exchanges (Rubin 2002). Because feelings of PSI deepen perceived intimacy, increase liking, and reduce feelings of uncertainty (Perse and Rubin 1989), viewers likely trust the mediated persona. This can reduce uncertainty and increase cooperation (Morgan and Hunt 1994), so that individuals are likely to reciprocate with personal disclosures and increased loyalty intentions (Porter and Donthu 2008). Therefore,

H2a. Feelings of PSI increase loyalty intentions.

H2b. Feelings of PSI increase willingness to provide information.

H3a. The positive impact of interactivity on the dependent variables is mediated by PSI.

H3b. The positive impact of openness on the dependent variables is mediated by PSI.

A multi-method approach is used to test these hypotheses. First, a survey methodology explores existing consumer–brand relationships on social media (Study 1). Second, an experimental study tests the hypotheses through altering the content of a fictitious company blog in order to manipulate perceived interactivity and openness in communication (Study 2). Lastly, boundary conditions are explored with a second experiment (Study 3).

Study 1: Survey

Design, Participants, and Procedure

The survey instrument aims to measure participants' relationships with brands through social media by reflecting on real encounters. It began by asking participants to "think about a brand, company, or service that you interact with using social media" and to "keep these interactions in mind while answering the survey questions." Participants named the brand and then briefly described their memory of the social media interaction. Participants were instructed to keep this memory in mind while completing the survey; therefore the reflection on past specific brand interactions was salient during the survey administration. Participants then answered questions relating to the constructs of interest (See Table 1), followed by demographic questions.

The 185 participants were recruited from an online panel (Amazon Mechanical Turk) and were paid for their participation in the study. Participation in the survey was limited to adults (over 17 years of age) living in the United States. Fifty-seven percent were women, and their average age was 27 years (SD = 10.06; range 18–69 years). Overall, the respondents were heavy Internet users; 50% of the sample reported using the Internet for two to four hours per day outside of work activities, 32% indicated more than four hours, and only 18% noted less than two hours. In terms of brand interactions, the vast majority of respondents (76%) interacted with 10 or fewer brands (43% 1–5 brands; 33% 6–10 brands), whereas only 24% did so with 11 or more. Types of brands varied widely and included retailers such as Victoria's Secret and Urban Outfitters to products such as Coca-Cola and Garnier. Platforms for interaction varied and included Facebook (78%), company websites or blogs (51%), Twitter (38%), e-mail (32%), YouTube (5%), Pinterest (4%), Foursquare (3%), and Google + (3%).

Measures

To test the hypotheses, constructs were captured using reflective, multi-item, seven-point Likert scales, anchored by "strongly disagree" (1) and "strongly agree" (7). The constructs rely on established scales from prior research in marketing and communications (see Table 1); some items were slightly altered to suit the study context.

Following the two-step procedure recommended by Anderson and Gerbing (1988), the measurement model was estimated prior to testing the relationships among the constructs in the research

Table 1
Study 1: constructs & psychometric properties of the measures.

Constructs	Scale item[a]	Factor loading	AVE[b]	ASV	CR	α
Perceived interactivity	[Brand] will talk back to me if I post a message.	.72	.63	.22	.83	.85
	[Brand] would respond to me quickly and efficiently.	.81				
	[Brand] allows me to communicate directly with it.	.83				
	[Brand] listens to what I have to say.	.73				
Openness	[Brand] is open in sharing information.	.83	.56	.25	.79	.79
	[Brand] keeps me well informed.	.77				
	[Brand] doesn't hold back information.	.64				
Parasocial interaction	[Brand] makes me feel comfortable, as if I am with a friend.	.71	.52	.43	.87	.83
	When I interact with [brand], I feel included.	.76				
	I can relate to [brand].	.70				
	I like hearing what [brand] has to say.	.71				
	I care about what happens to [brand].	.75				
	I hope [brand] can achieve its goals.	.71				
Willingness to provide information	I'm willing to provide information about myself to [brand].	.71	.54	.31	.78	.77
	I'm happy to provide information about my needs to [brand].	.79				
	I'm willing to complete a survey for [brand].	.71				
Loyalty intentions	I'm willing to say positive things about [brand] to others.	.92	.60	.27	.82	.75
	I'm willing to encourage close others to do business with [brand].	.80				
	I plan to do business with [brand] in the next few years.	.57				

[a] The perceived interactivity items were adapted from McMillan and Hwang (2002), Song and Zinkhan (2008), Thorson and Rodgers (2006); openness in communication was adapted from the opportunism scale by John (1984) as well as the communication scale from Anderson and Weitz (1992); loyalty intentions reflect items by Zeithaml, Berry, and Parasuraman (1996); willingness to provide information items are from Schoenbachler and Gordon (2002); and the parasocial interaction measures came from Rubin, Perse, and Powell (1985).

[b] CR = Composite Reliability; AVE = Average Variance Extracted; ASV = Average Shared Squared Variance; α = Cronbach's alpha.

model. The measurement model consisted of 19 measurement items and five factors. Overall, model fit was acceptable ($\chi^2(142) = 343.95$ $p = .000$; CFI = .97; IFI = .97; SRMR = .07; RMSEA = .09[1]). All items loaded significantly on their respective constructs, in support of the convergent validity of the measurement items. The internal consistency estimates (composite reliability) and amount of variance extracted for each construct in relation to measurement error met acceptable threshold levels (Fornell and Larcker 1981; Nunnally and Bernstein 1994). As shown in Table 1, the composite reliability coefficients all exceeded the .60 standard (Bagozzi and Yi 1988), and the average variance extracted (AVE) for all of the construct measures exceeded Fornell and Larcker's (1981) .50 benchmark. In addition, evidence of satisfactory discriminant validly is indicated by the AVE exceeding the average shared squared variance (ASV) (Fornell and Larcker 1981; Hair et al. 2010).

Analysis and Results

Structural equation modeling (SEM), with LISREL 8.8 software, was used to test the hypothesized relationships. Separate structural models were run for each of the two independent variables (interactivity and openness) since the introduction of a second independent variable as an antecedent to a mediating variable in an SEM mediation analysis alters the focal mediation path coefficients making them no longer invariant (Iacobucci, Saldanha, and Deng 2007).[2] In both models, loyalty and willingness to provide information served as the dependent variables, and PSI served as the mediating variable. The structural models exhibited acceptable fit (Openness: χ^2 (85) = 206.81, $p = .000$; CFI = .95; IFI = .95; SRMR = .07; RMSEA = .08; Interactivity: χ^2 (99) = 281.08, $p = .000$; CFI = .95; IFI = .95; SRMR = .07; RMSEA = .10[1]).

Antecedents and Consequences of PSI

In support of H1a and H1b, results (See Fig. 1) show a positive and significant direct effect of openness and interactivity on PSI (path a). Consumers' perceptions of interactivity and openness thus increase their feelings of PSI. In support of H2a and H2b, the effects of PSI on loyalty and willingness to provide information are both positive and significant (path b). Therefore, PSI increases loyalty intentions and willingness to provide information.

Mediation

The results (See Fig. 1) show that the total effect of openness on loyalty ($c' = .43$, $p < .05$) and the total effect of openness on willingness to provide information ($c' = .38$, $p < .05$) are both

[1] Recent evidence (Chen et al. 2008) claims that the arbitrary .05 cut-off point of RMSEA rejects too many valid models with small samples, as RMSEA is sensitive to sample size. These researchers suggest using multiple indicators to determine model fit.

[2] The results of a single structural model that included both independent variables as predictors of the mediating variable produced identical results in terms of path significance, but slight differences in the path coefficients.

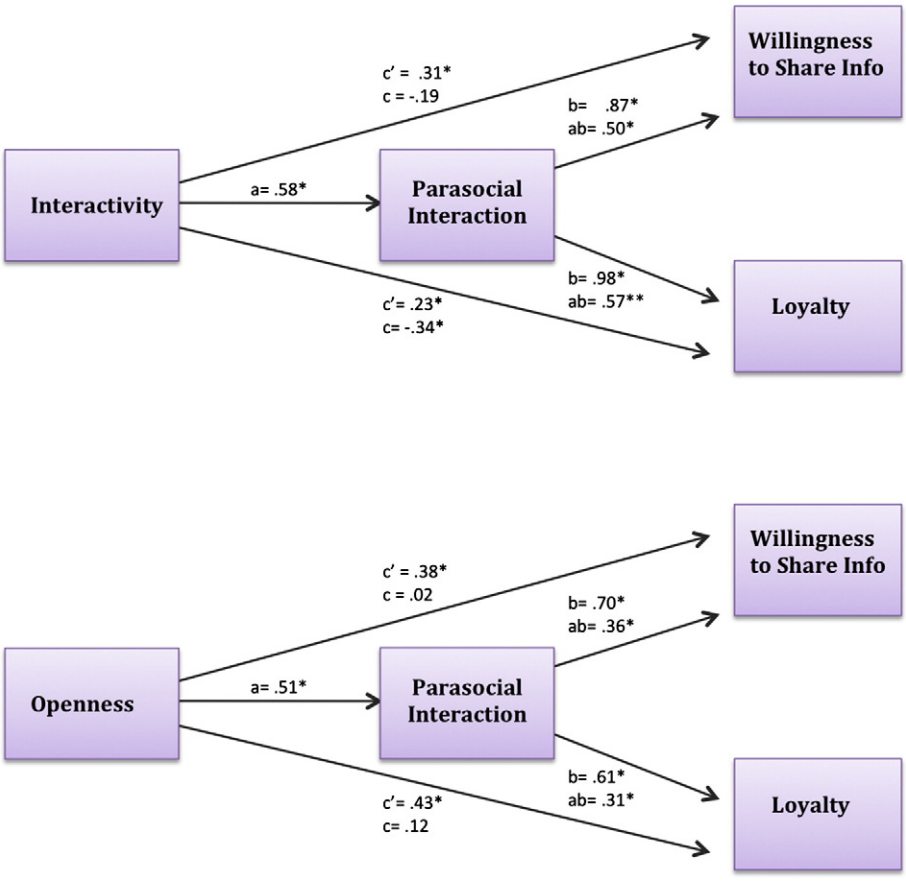


Fig. 1. Study 1: Mediation Analysis — Structural Equation Modeling with LISREL. Notes: path a = direct effect of the independent variable on the mediating variable; path b = direct effect of the mediating variable on the dependent variable; path c = direct effect of the independent variable on the dependent variable; path c′ = total effects (independent variable on the dependent variable accounting for the mediator); path ab = indirect effect of the independent variable on the dependent variable through the proposed mediator. * = p < .05; ** = p < .001.

positive and significant. The direct effect of openness is not significant for either loyalty (c = .12, $p > .05$) or willingness to provide information (c = .02, $p > .05$). The indirect effect of openness through PSI is positive and significant for loyalty (ab = .31, $p < .05$) and willingness to provide information (ab = .36, $p < .05$). Taken together, these results provide evidence of PSI's full mediation for the effect of openness on the dependent variables.

The results for interactivity show that the total effect of interactivity on loyalty (c′ = .23, $p < .05$) and the total effect of interactivity on willingness to provide information are both positive and significant (c′ = .31, $p < .05$). The direct effect of interactivity is not significant for willingness to provide information (c = −.19, $p > .05$); however the direct effect is significant for loyalty (c = −.34, $p < .05$). The indirect effect of interactivity through PSI is positive and significant for loyalty (ab = .57, $p < .001$) and for willingness to provide information (ab = .50, $p < .05$). Taken together, these results provide evidence of full mediation for the effects of interactivity on willingness to provide information and evidence of competitive (partial) mediation for the effects of interactivity on loyalty (Zhao, Lynch, and Chen 2010). These results provide support for H3a and H3b.

Discussion

This study, using survey methodology and SEM mediation analysis, offers insights into the role that PSI plays in the relationship between social media message cues and the focal outcome variables. That is, brands can create a sense of PSI through message cues that signal interactivity, as well as openness in communication. This sense of feeling connected with the brand through the interaction (PSI) goes beyond the interaction itself and drives increased feelings of loyalty intentions and willingness to provide information to the brand.

These effects help clarify the role of PSI in developing consumer–brand relationships and provide evidence of mediation; however, while this methodology can be considered superior for exploring correlation relationships and for assessing meditation (Iacobucci, Saldanha, and Deng 2007), it fails to offer evidence to support causality. Therefore, an experimental

methodology was undertaken to offer causal support, as this methodology offers superior evidence for identifying causal relationships (Iacobucci, Saldanha, and Deng 2007). The experiment was designed to investigate whether PSI can be developed through the characteristics and content of a brand's message and also tests whether feelings of PSI can be fostered through a single interaction.

Study 2: Experimental Manipulation of PSI Through Interactivity and Openness

Design, Participants, and Procedure

The experiment followed a single factor (PSI: low vs. high) between-subjects design. The 66 participants were recruited from the same online panel used for Study1 and were paid for their participation in the study. Data were screened to ensure unique respondents for both studies. As in Study 1, participation in the survey was limited to adults (over 17 years of age) living in the United States. They were 52% women, and their average age was 31 years (SD = 10.50; Min = 18, Max = 69). In terms of Internet usage, 33.3% reported use of five or more hours per day, 18.2% for four-five hours per day, 15.2% for three-four hours per day, 21.2% for two to three hours per day, and 12.1% for two hours or less. Furthermore, 74.2% of participants reported spending more than 1 h per day on social media sites and all reported to engage with brands on these platforms (56.1% reported actively engaging with 1–5 brands, 21.2% reported actively engaging with 6–10 brands, 22.7% reported actively engaging with more than 10 brands).

For this experiment, a fictitious website was created by a professional web developer to serve as the vehicle to test the antecedents and outcomes of PSI. In order to ensure realism and familiarity, the site's aesthetics and functionality were based upon a visual survey of corporate blogs from the same product category (fashion retail). The site followed a typical design, where users' comments were displayed on the page in chronological order from top to bottom (most recent) and could be identified by screen name and avatar. At the start of the study, participants were told that they would interact with a brand on this site and then answer questions about the interaction.

Prior to interacting with the fictitious brand, participants created an account on the site by selecting a screen name and an avatar, to which they could add personalization. Following this step, they read a description of a fictitious clothing company, Lemon Federation. The narrative described the retailer as specializing in offering the latest styles at reasonable prices. The description asserted that there were limited stores currently open in the United States, but the company was planning a national expansion. Following the account creation, participants were randomly assigned to one of the experimental conditions and directed to the blog.

PSI Manipulation

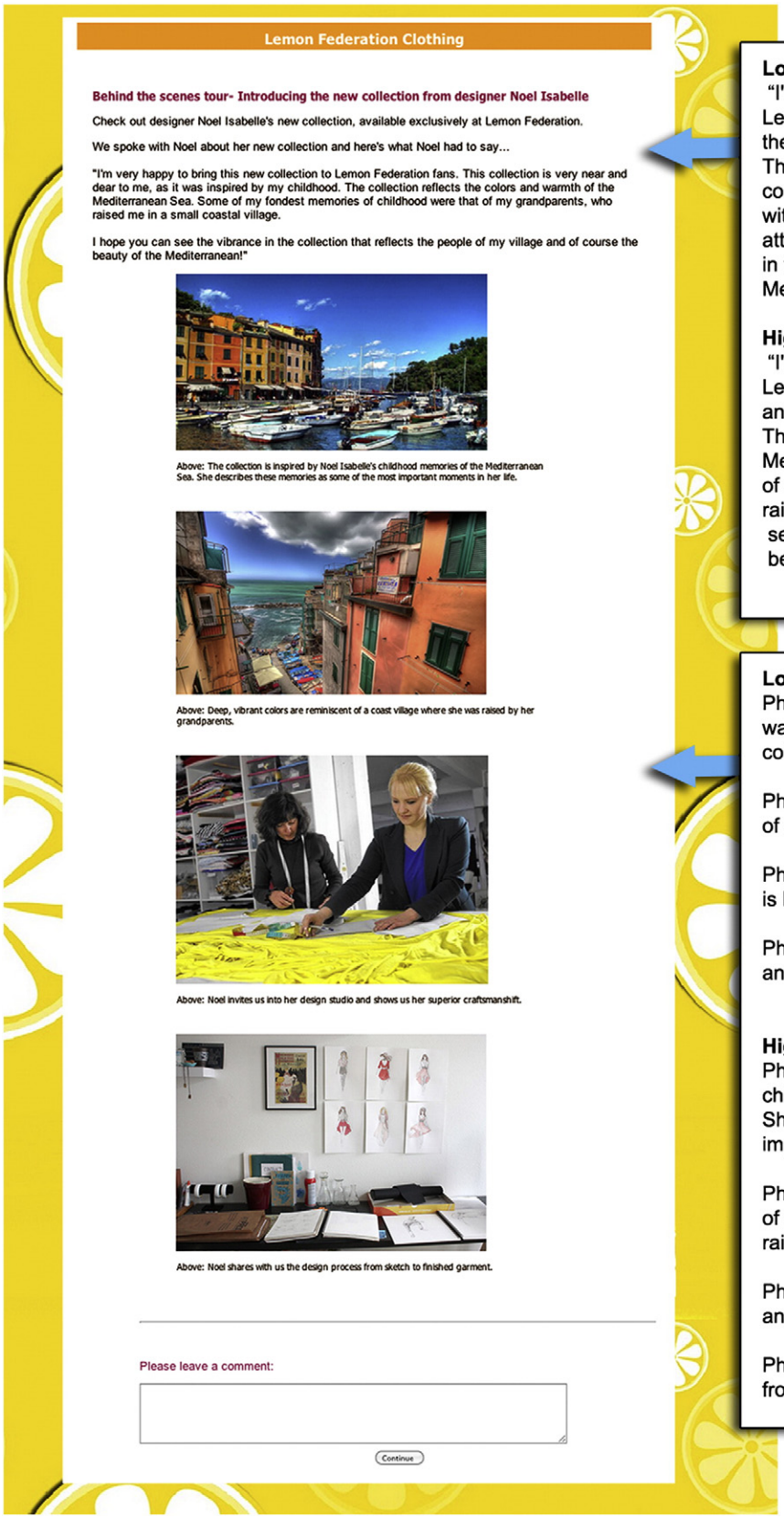
Study 1 provided evidence that PSI can be formed through the use of two message cues: interactivity and openness in communication; therefore these two cues were used to create the high vs. low PSI conditions. The manipulation for openness in communication was established through the content of the blog post. Content displayed on both the high and low openness conditions appeared visually equal (i.e., both had the same images, colors, formats, and content length); the sole difference resided in the text (see Fig. 2). This assured equality in terms of visual presentation. Pages for both high and low conditions presented a story about a new collection from a Lemon Federation designer who described the Mediterranean Sea as being the inspiration for her new line. While both high and low openness conditions described the Mediterranean Sea as the designer's inspiration for the collection, in the high openness condition the sea was linked to the designer's childhood memories. This personal connection was absent in the low openness condition (see Fig. 2). Furthermore, the image captions in the high openness condition highlighted this personal connection and reinforced that viewers were getting a "behind the scenes" tour of the design studio.

At the end of the blog post, participants were given the opportunity to post a comment, which is a common blog capability. All participants completed this task and were included in the analysis; their comments indicated that they took the task seriously, including detailed messages to the brand and average responses longer than one sentence.

After posting a comment, participants were given the option to return to the blog in order to read others' comments. Participants were directed to the same blog post, which now contained a series of comments. Their comment was displayed along with their chosen avatar and screen name, amidst comments from the brand and other users (see Fig. 3). Interactivity was manipulated through message characteristics of the comments. Specifically, this was achieved by modifying response time and personalization of the message. Both the response speed and personalization of the message manipulations were adapted from Song and Zinkhan's (2008) work on perceived interactivity.

In the high PSI condition (n = 33), the response was personalized (i.e., the participants were directly addressed by screen name) in one of the brand's comments (Song and Zinkhan 2008). Moreover, this acknowledgement came directly after the participant's comment, indicating a timely response (Song and Zinkhan 2008). Other user comments and promotional comments from the brand were included on the page following these two comments to assure equivalence between the conditions. These comments were modeled from real comments found on similar clothing retailers' Facebook pages and blogs.

In the low PSI conditions (n = 33), promotion-focused comments from the brand and others' comments followed the participant's comment. Here, the participant was never directly addressed in a comment; instead the brand just gave a generic comment to readers that began with the word "Fans". The brand's generic response was placed after a promotional comment by the brand amidst other comments, indicating a less timely response. After reading the comments page, participants responded to a series of questions about their interaction with Lemon Federation, including the dependent and independent variable items and demographics.



Low Condition Opening Vignette:
"I'm very happy to bring this new collection to Lemon Federation fans. This collection reflects the colors and warmth of the Mediterranean Sea. The deep colors of the ocean and warm Earth tones come alive in these clothes. Each piece is made with the finest quality materials and with incredible attention to detail. I hope you can see the vibrancy in the collection that reflects the beauty of the Mediterranean!"

High Condition Opening Vignette:
"I'm very happy to bring this new collection to Lemon Federation fans. This collection is very near and dear to me, as it was inspired by my childhood. The collection reflects the colors and warmth of the Mediterranean Sea. Some of my fondest memories of childhood were that of my grandparents, who raised me in a small coastal village. I hope you can see the vibrancy in the collection that reflects beauty of my village and the Mediterranean!"

Low Condition Photo Captions:
Photo 1: The collection is inspired by the colors and warmth of the Mediterranean Sea.These vibrant colors are found throughout the collection.

Photo 2: Deep, vibrant colors are reminiscent of a coast village town by the Mediterranean Sea.

Photo 3: Like all of our clothing, this collection is handmade with superior craftsmanship.

Photo 4: The design process begins with a sketch and results in a finished garment.

High Condition Photo Captions:
Photo 1: The collection is inspired by Noel Isabelle's childhoood memories of the Mediterranean Sea. She describes these memories as some of the most important in her life.

Photo 2: Deep, vibrant colors are reminiscent of a coast village where Noel's grandparents raised her.

Photo 3: Noel invites us into her design studio and shows us her superior craftsmanship.

Photo 4: Noel shares with us the design process from sketch to finished garment.

Fig. 2. Study 2: openness manipulation, blog post.

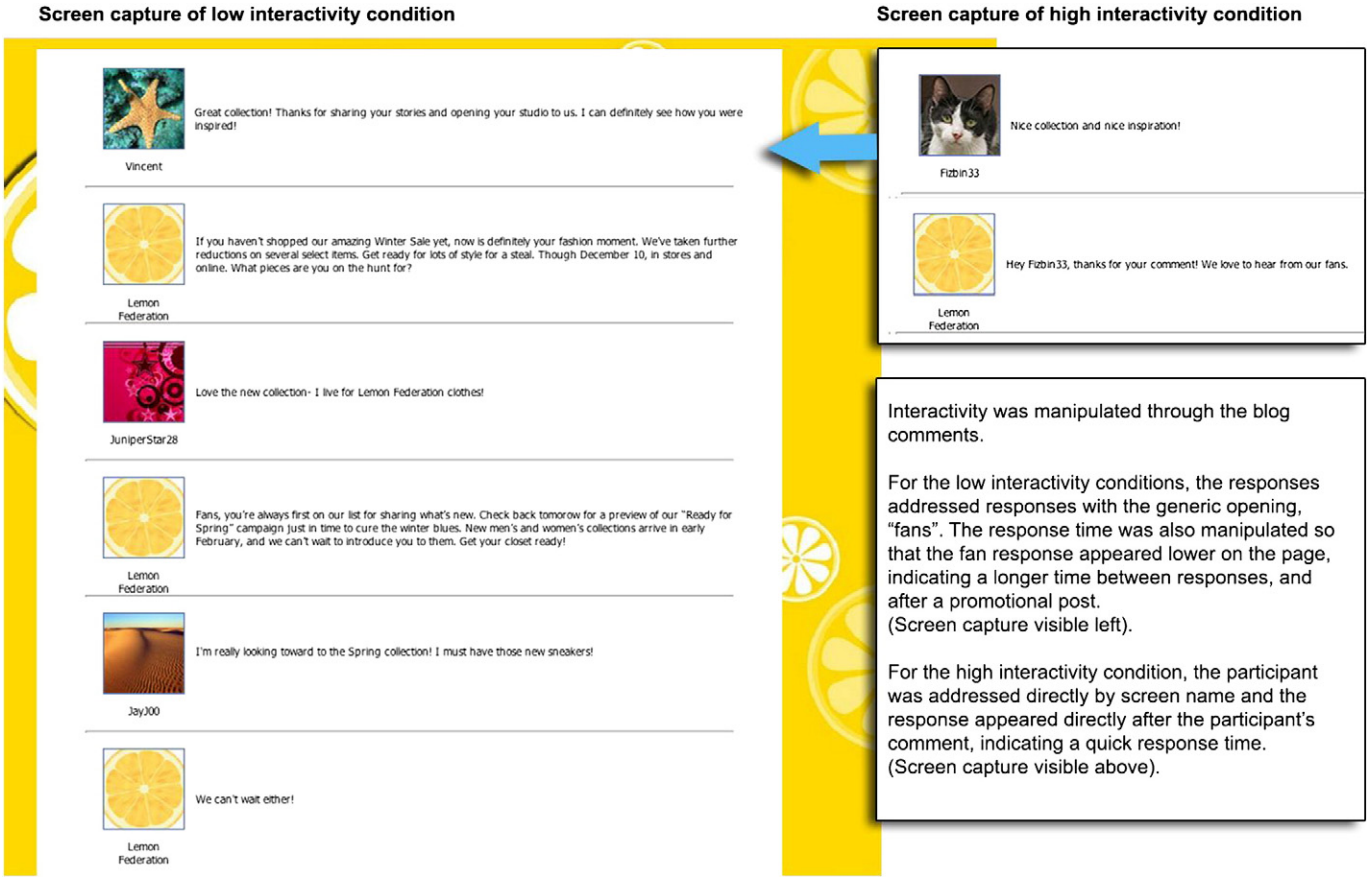


Fig. 3. Study 2: interactivity manipulation, sample comments page.

Measures

The same measures from Study 1 were collected in Study 2, using 9-point Likert scales. The coefficient alphas computed for the five dimensions (PSI $\alpha = .95$; Loyalty $\alpha = .93$; Willingness to Provide Information $\alpha = .82$; Interactivity $\alpha = .89$; Openness $\alpha = .80$) indicate high internal consistency.

Analysis and Results

Manipulation Checks

ANOVAs provided evidence that the manipulations worked as intended. Participants in the low PSI condition reported significantly lower perceptions of PSI than those in the high condition ($M_{low} = 6.05$; $M_{high} = 7.18$; $F_{1,64} = 7.56$, $p < .05$). Likewise, those in the low PSI condition reported lower interactivity ($M_{low} = 6.45$; $M_{high} = 7.88$; $F_{1,64} = 15.48$, $p < .05$) and openness ($M_{low} = 5.60$; $M_{high} = 6.46$; $F_{1,64} = 3.91$, $p = .05$) than those in the high PSI condition.

Outcomes

A MANOVA, with loyalty intentions and willingness to provide information as the dependent variables and level of manipulated PSI (condition) as the independent variable, revealed a significant relationship between conditions and the dependent variables (Wilks' $\lambda = .84$, $F_{2,63} = 5.99$, $p < .05$, partial $\eta^2 = 16$). Specifically, the results highlighted significant differences between conditions for loyalty intentions ($F_{1,64} = 8.95$, $p < .05$, partial $\eta^2 = .12$) and willingness to provide information ($F_{1,64} = 5.43$, $p < .05$, partial $\eta^2 = .15$). The means in the low PSI condition (n = 33) were significantly lower than those in the high (n = 33) PSI condition for both willingness to provide information ($M_{low} = 5.42$, $M_{high} = 6.83$, $p < .05$) and loyalty ($M_{low} = 5.74$, $M_{high} = 6.92$, $p < .05$). Taken together, these results provide support for H2.

Mediation Analysis

A bootstrap mediation analysis (Preacher and Hayes 2004, 2008; Zhao, Lynch, and Chen 2010) tested the hypothesized relationships in the research model. The analysis estimates relied on 5000 bootstrap samples. According to Preacher and Hayes (2004, 2008) and Zhao, Lynch, and Chen (2010), full mediation occurs when a non-significant direct path from the independent variable to the dependent variable (c) is present simultaneously with a significant indirect path (ab). Partial mediation occurs when both the indirect (ab) and direct (c) paths are significant.

Parasocial interaction fully mediated the relationship from openness to both the outcome variables. The path from openness to loyalty revealed a significant indirect effect (ab = .53, 95%

confidence interval = .32, .77) and a non-significant direct effect (c = .06, t = .51, p = .61); the path from openness to willingness to provide information included a significant indirect effect (ab = .42, 95% confidence interval = .20, .67) and a non-significant direct effect (c = .23, t = 1.79, p = .08).

In terms of interactivity, PSI was found to fully mediate the relationship between interactivity and loyalty with a significant indirect effect (ab = .52; 95% confidence interval = .22, .82) and a non-significant direct effect (c = .23, t = 1.60, p = .11). However, PSI only partially mediated the influence of interactivity on willingness to provide information due to a significant indirect effect (ab = .30, 95% confidence interval = .01, .66) and a significant direct effect between interactivity and willingness to provide information (c = .54, t = 3.30, p = .002).

These results, in support of H3a and H3b, provide evidence that the positive effect of communication characteristics can be explained by feelings of PSI. This mediation analysis also supports findings from Study 1. Similar to Study 1, evidence of full mediation was found for the relationships between openness and the dependent variables. While Study 1 found support for partial mediation between interactivity and loyalty and full mediation between interactivity and willingness to provide information, this analysis supports full mediation for interactivity and loyalty and partial mediation between interactivity and willingness to provide information. Taken together, these results offer support that mediation is stronger for openness than interactivity.

Discussion

Feelings of PSI, created through perceived interactivity and openness in communication, relate positively to feelings of loyalty intentions and willingness to provide information. These results further support Study 1's findings and demonstrate that it is possible to increase PSI through message cues. In terms of message cues, perceptions of interactivity can be driven by timely responses and by directly addressing users by name in response to their comments, which can further increase PSI levels. Furthermore, PSI can be fostered through openness in communication, such as sharing seemingly personal details, and establishing feelings of a one-to-one relationship. The resulting sense of intimacy cultivated through PSI can strengthen the relationship between the user and the brand, resulting in increased willingness to provide information to the brand and strengthening of loyalty intentions.

The high PSI manipulation used in this study represents a personalized, direct response. Arguably, while a response of this nature may be driven by a mediated conversation with another person, this type of response can also be achieved through sophisticated automation software. The high PSI condition resulted in higher feelings of PSI, which in turn led to higher loyalty intentions and willingness to provide information than the control condition. The results of this study assume that individuals receiving these messages are unaware of the possibility that the responses may be automated; however, could consumer knowledge of computer automation capabilities affect the ability for these message cues to foster PSI? Study 3 is designed to answer this question.

Study 3: Impact of Computer Automation Salience on PSI

The experiment was designed to investigate whether knowledge of computer response automation can affect the ability to foster PSI through the message cues described in Studies 1 and 2. Being aware of the possibility that the reply from the mediated other may stem from an automated response should dissolve the perception that the individual is engaged in a direct two-way conversation with the mediated other. Therefore, the illusion that the mediated other is talking directly to him or her, and feelings of PSI should dissipate.

Design, Participants, and Procedure

The experiment followed a 2 (PSI: low vs. high) × 2 (automation salience: absent vs. present) between-subjects design. The 129 participants were recruited from the same online panel used in the previous studies and were paid for their participation in the study. Data were screened to ensure unique respondents for all studies. As described previously, participation was limited to adults (over 17 years of age) living in the United States. They were 57% women, and their average age was 31 years (SD = 10.14; Min = 18, Max = 72). In terms of Internet usage, 35.7% reported more than five hours of use per day, 13.2% for four–five hours per day, 19.4% for three–four hours per day, 14.0% for two to three hours per day, and 17.9% for two hours or less. Furthermore, 55.8% of participants report spending more than 1 h per day on social media sites and all reported engaging with brands on these platforms (72.1% reported actively engaging with 1–5 brands, 17.1% reported actively engaging with 6–10 brands, 10.8% reported actively engaging with more than 10 brands).

This experiment utilized the same fictitious website described in Study 2. Participants in the automation salience absent conditions followed the exact same procedure outlined in Study 2 (i.e., created an account, chose an avatar, etc.) and were exposed to the same PSI manipulations (high vs. low) described in Study 2. Those in the automation salience present conditions also followed the same procedure and task outlined in Study 2 with one exception; they were first assigned a short reading task prior to completing the main task. This short story, adapted from a news story appearing on Mashable.com, described how advances in technology are allowing robots to replace human tasks, specifically in areas such as food preparation and social media responses (see Appendix A). After completing this brief reading task, respondents were asked to rate their enjoyment of the article with two 5-pt Likert scale questions (I enjoyed reading the article; the article was interesting). These questions were distractor questions and not related to the research. Following the completion of these two questions, participants were directed to "Part 2" of the study, which was described as being unrelated to the previous task.

Measures

The same items reported in Study 2 were used to measure PSI, Loyalty, Willingness to Provide Information, Interactivity, and Openness, and were collected using 9-point Likert scales. In addition, all participants completed a subset of 8 items from the Computer Attitude Scale (Nickell and Pinto 1986) and a scale measuring Consumer Skepticism (Koslow 2000; Sääksjärvi and Morel 2010) (see Appendix B for measures) after completing the other measures related to PSI. The coefficient alphas computed for the seven dimensions (PSI $\alpha = .96$; Loyalty $\alpha = .94$; Willingness to Provide Information $\alpha = .89$; Interactivity $\alpha = .91$; Openness $\alpha = .79$; Computer Attitude Scale $\alpha = .80$; Consumer Skepticism $\alpha = .70$) indicate high internal consistency.

Analysis and Results

Manipulation Checks

ANOVAs provide evidence that the PSI manipulations worked as intended. Participants in the low (n = 64) PSI conditions reported significantly lower perceptions of PSI than those in the high (n = 65) conditions ($M_{low} = 5.93$; $M_{high} = 6.71$; $F_{1,127} = 6.06$, $p < .05$). Likewise, those in the low PSI conditions reported lower interactivity ($M_{low} = 4.69$; $M_{high} = 5.33$; $F_{1,127} = 7.49$, $p < .05$) and openness ($M_{low} = 5.53$; $M_{high} = 6.05$; $F_{1,127} = 3.182$, $p = .07$) than in those the high PSI conditions.

Outcomes

A MANOVA, with loyalty intentions and willingness to provide information as the dependent variables and condition as the independent variable, revealed a significant relationship between manipulated PSI conditions and the dependent variables (Wilks' $\lambda = .89$, $F_{6,248} = 2.378$, $p < .05$, partial $\eta^2 = .06$). Planned contrasts highlight significant differences between conditions.

In the absence of automation salience (i.e., participants who were not exposed to the story on computer automation) the results mirrored those found in Study 2. The means in the low PSI condition (n = 33) were significantly lower than those in the high (n = 34) PSI condition for both willingness to provide information ($M_{low} = 5.69$, $M_{high} = 6.85$, $p < .05$) and loyalty ($M_{low} = 5.85$, $M_{high} = 7.27$, $p < .05$). ANOVAs provide evidence that the PSI manipulations worked as intended. Additionally, significant differences were found across participants in the low versus high PSI conditions ($M_{low} = 5.89$; $M_{high} = 7.22$; $F_{1,67} = 14.37$, $p < .05$).

Conversely, the presence of automation salience (i.e., participants who read the story on computer automation) revealed no significant differences across PSI conditions for the dependent variables. In these conditions, the means in the low PSI condition (n = 30) were not significantly different than those in the high (n = 30) PSI condition for both willingness to provide information ($M_{low} = 5.79$, $M_{high} = 5.76$, $p = .94$) and loyalty ($M_{low} = 6.17$, $M_{high} = 5.96$, $p = .65$). Here no significant differences were found in terms of PSI ($M_{low} = 5.98$; $M_{high} = 6.11$; $F_{1,58} = .06$, $p = .81$).

Exposure to the story on computer automation (automation salience present condition) did result in higher scores on the Computer Attitude Scale ($M_{present} = 5.82$, $M_{absent} = 4.49$, $p > .01$) and no significant differences were found when comparing within these conditions in terms of the PSI manipulation ($M_{present/low\ PSI} = 5.81$ vs. $M_{present/high\ PSI} = 5.82$; $M_{absent/low\ PSI} = 4.57$ vs. $M_{absent/high\ PSI} = 4.42$).

A final ANOVA was run with skepticism as the dependent variable and condition as the independent variable in order to rule out the possibility that the observed differences between the conditions were due to variations on individual differences on this trait. The results reveal no significant differences across conditions ($F_{3,125} = .76$, $p = .52$).

Mediation Analysis

As in Study 2, a bootstrap analysis with 5000 resamples (Preacher and Hayes 2004, 2008) was used for mediation testing. Parasocial interaction fully mediated both paths from openness to the outcome variables. The path from openness to loyalty revealed a significant indirect effect (ab = .64, 95% confidence interval = .51, .79) and a non-significant direct effect (c = .04, $t = .63$, $p = .53$); the path from openness to willingness to provide information included a significant indirect effect (ab = .59, 95% confidence interval = .45, .76) and a non-significant direct effect (c = .09, $t = 1.24$, $p = .22$).

In terms of interactivity, PSI was found to fully mediate both paths from interactivity to the outcome variables. The path from interactivity to loyalty revealed a significant indirect effect (ab = .86, 95% confidence interval = .67, 1.06) and a non-significant direct effect (c = .10, $t = 1.12$, $p = .26$); the path from interactivity to willingness to provide information included a significant indirect effect (ab = .80, 95% confidence interval = .59, 1.02) and a non-significant direct effect (c = .14, $t = 1.35$, $p = .18$).

Discussion

Study 3 created a scenario where for some participants, the use of computer automation for human tasks, such as social media response, was made salient. The use of computer automation was made salient by asking participants to read a news story about how technology is increasingly being used to replace human tasks. Interestingly, this reading task resulted in an increased score on the Computer Attitude Scale, which reflects a more general positive attitude towards computers. Yet, although participants exhibited more positive feelings towards computers, the awareness that a computer, not a human being, may be behind the interaction dampened the ability for message cues to foster PSI.

General Discussion and Conclusions

These three studies together provide evidence that parasocial interaction theory can be helpful in understanding how consumer–brand relationships are established through social media. Study 1 provides evidence in support of this relationship by examining the antecedents and outcomes of PSI using a survey methodology. The Study 1 analysis also provides evidence of full mediation for the effect of openness on willingness to provide

information and loyalty and for the effect of interactivity on willingness to provide information. Support for competitive mediation (Zhao, Lynch, and Chen 2010), is found for the effect of interactivity on loyalty intentions. Study 2 adopts an experimental methodology by altering brand message cues to induce PSI. The combined results of these studies provide strong evidence of causality and enhance generalizability. Study 3 reveals that the PSI effects may not hold when the possibility that the brand's social media response may be automated becomes salient.

Theoretical Contributions

This research provides a theory for understanding the processes underlying the development of consumer–brand relationships in social media environments. As social media use continues to increase and marketers turn their attention to investing in such channels, understanding the psychological underpinnings of customer relationships becomes increasingly vital. With its multi-method approach, the current research offers a theoretical explanation for how consumer–brand relationships develop in these environments.

Specifically, this research establishes openness in communication and perceived interactivity as antecedents of PSI. Moreover, in terms of the outcomes of PSI, the structural equation modeling analysis and experiments find a positive relationship with desirable relationship variables, loyalty intentions and willingness to provide information. Furthermore, mediation analysis reveals that the positive effects created by interactivity and openness can be partially explained by PSI. Brands can create a sense of PSI through crafting messages to include elements that signal that the brand is listening and responding to customers and by creating content that expresses openness in communication. Ultimately, the sense of feeling connected to the brand through the interaction, not merely the interaction itself, drives these outcomes.

This research also further extends the PSI literature from the field of communications to marketing. Parasocial interaction theory was first proposed more than 50 years ago as means to understand how people interacted with personas in mass media, such as radio and television (Horton and Wohl 1956). The bidirectional communication capability of the Internet creates another ideal platform for generating feelings of PSI. This feature, in conjunction with other distinguishing traits of the Internet, such as its 24-h access, arguably provides an even richer medium for building and strengthening consumer–brand relationships.

Managerial Implications

Social media use is exploding and online channels have become essential platforms for marketing. Yet little academic research is available to help marketers understand the best practices for building relationships with consumers through such channels. The findings of this research suggest some guidelines for engaging with consumers though. Specifically, through carefully designed message content and message cues, marketers can foster a sense of PSI.

While the Internet offers marketers the ability for direct two-way communications with consumers, the increasing consumer-brand activity on social media platforms may make it impossible for direct individual responses. New technologies to help marketers manage consumer-brand interactions, such as software that allows personalized automated responses based on variables such as message content and user location (Defren 2012; Owyang 2012a, 2012b; Zebida 2012), are on the rise. As marketers decide to move to these technologies for social media engagement, they need to be aware of the potential limitations and perils that come with automation (Eridon 2011).

This research provides evidence that even automated responses, such as the ones used in Studies 2 and 3, are capable of fostering feelings of a personal interaction between the consumer and the brand through crafted message cues that enhance the perception of a one-to-one interaction, as long as consumers are unaware that the response is driven by a pre-programmed script. Arguably, a one-to-one response from a brand representative is ideal, but realistically this may not be possible, especially for brands with heavy social media activity. Furthermore, message content can also be designed to facilitate feelings of PSI. As highlighted in Study 2, providing personal stories and "back-stage" details, created the sense of open communication between the brand and its customers. The manipulation in Study 2 illustrates that integrating personal details to a promotional message can yield a positive impact on both PSI and the relationship outcome variables.

Moreover, this research provides support for social media expenditures, something that marketers have struggled to define in terms of the return on investment. By establishing PSI with consumers, companies strengthen their relationships and increase loyalty intentions and willingness to provide information. As the world continues to become transparent due to the open nature of the Internet, it is essential that marketers realize the potential pitfalls and opportunities of these channels. Understanding the underpinnings of the relationships they create with customers will be the key.

Limitations and Directions for Further Research

The two antecedents to PSI, perceived interactivity and openness, examined herein significantly increase PSI; yet, other antecedents also are likely. These message cues were chosen due to their relationship with techniques used in traditional PSI environments. Future research should examine other antecedents that may be unique to this medium. This research also included only two outcome variables; therefore, future research may examine further consequences of PSI in social media communications. Furthermore, future research may examine if PSI contributes over and above other established marketing constructs, such as trust.

One can argue that employees responding on behalf of the brand are linked to the brand and therefore messages are connected to the brand; however many corporate social media accounts are managed by outside firms, which remove direct brand connections. Moreover, as previously mentioned, the development and adoption of automated social media engagement software have

decreased the number of truly one-to-one human responses. This technology allows for a continuum of response options from computer assisted human response (such as Hootsuite's ability to choose from sets of preapproved messages that can be sent with the touch of a button) to full automated responses that can be triggered by intelligent algorithms without human assistance. Further research exploring different levels of human to computer responses is warranted.

Study 3 provided evidence supporting the notion that the ability for message cues to foster perceptions of PSI is attenuated when users are aware of the use of computer automated social media responses. In this study, salience was achieved through a reading task; however, it's likely that other message cues can trigger this. Future researchers may also choose to investigate this. Also, while individual differences in terms of skepticism were examined here, future researchers may consider other individual traits such as Need for Cognition. It would be interesting to investigate other differences between users. For instance, the antecedents and outcomes of PSI vary for different age groups. Perhaps older users of social media might not be as trusting as younger users.

This research has taken the first step to establish that social media environments are capable of producing PSI. The unique technological capabilities of the Internet provide an ideal environment for developing PSI, arguably more so than previous communication mediums. Future research can expand on this work to consider developing a new scale for measuring PSI in this dynamic environment.

The experimental studies (Studies 2 and 3) manipulated perceived interactivity and openness to create high and low PSI conditions; however, this design does not allow for the investigation of interaction effects and the relationship between interactivity and openness. For instance, a high interactive situation may also involve revealing identity-related information and disclosing personal information may raise the level of perceived interactivity in addition to openness. Additional research may help further clarify the relationships among these constructs as this research does not concretely answer the question of whether or not both interactivity and openness are needed for creating high PSI experiences. The results of the three studies partly illuminate this issue as all three found full mediation for openness, but Study 1 and Study 2 only showed partial mediation for interactivity.

Lastly, the experimental studies focused solely on a fictional brand's blog; yet, the results parallel those found in Study 1, which reported respondents' interactions with real brands on various social media platforms. Although most of the respondents reported interacting with brands on similar sites, an investigation across other may be warranted given evidence of varying content across platforms (Smith, Fischer, and Yongjian 2012).

Appendix A. Study 3 Manipulation

INSTRUCTIONS: *Please read the following article, which was published in The Atlantic and reported by Mashable (a leading technology blog) last fall. After reading the article you will be asked some questions.*

Can a Robot Learn to Cook?

The art of the perfect chicken soup comes from hands-on experience and social interaction. If robots master that, what separates them from us?

Everyone's coming over to watch the big game. You've got beer, a giant high-definition television, and a well-deserved reputation for serving wings hotter than Dante's eighth circle of hell. Unfortunately, you are pressed for time. Wouldn't it be great if a machine like Rosey from The Jetsons could quickly prepare them? Maybe you could even pass off the dish as your own!

Then again, maybe not. Would Rosey's version taste like yours, or would her rendition expose your duplicity? Would Rosey know when the chicken pieces hit the ideal state of crispiness without being raw inside? Most importantly, could she discern when the spice Rubicon was crossed?

As every cook knows, mechanically following a recipe will only take you so far. Nevertheless, Gary McMurray, chief of the Food Processing Division of Georgia Tech's Research Institute, believes robots will acquire the knowledge needed to debone and butcher a chicken through the support of custom algorithms underwritten by complex mathematical equations.

For the sake of argument, let's say McMurray is right and it is only a matter of time before technology cuts chickens with the same speed, dexterity and accuracy as humans. Indeed, glimpses of the future are already here. Robots have already taken over a number of "human jobs" – For instance, many companies are increasingly using robots to respond to posts and questions on social media and the Chinese have developed noodle-bots that can hand-slice noodles into pots of boiling water. But will technology ever replicate the deft touch of China's best noodle pullers? Or completely replace human responses on social media?

Appendix B. Computer Attitude Scale and Consumer Skepticism Constructs (Study 3)

Computer Attitude Scale (items from Nickell and Pinto 1986)

- Computers can eliminate a lot of tedious work for people.
- The use of computers is enhancing our standard of living.
- Computers are dehumanizing to society.*
- There are unlimited possibilities of computer applications that haven't even been thought of yet.
- Computers turn people into just another number.*
- Computers are lessening the importance of too many jobs now done by humans.*
- Computers are bringing us into a bright new era.
- Soon our world will be completely run by computers.*

Notes: *reverse scored; Higher score indicates more favorable attitude towards computers.

Consumer Skepticism Scale (items from Sääksjärvi and Morel; originally adapted from Koslow 2000)

- I tend to question information.
- Before accepting anything from others, I first need to have critically reflected on it myself.
- My attitude in life is: seeing is believing.
- I am suspicious by nature.
- My friends and acquaintances think I am a skeptic.

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